ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2024 P2H Worked Solutions

Jun 2024 | P2H

31 questions ▪ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2024 P2H - Jun 2024 | P2H

1 Here are eight numbers written in order of size

$$h \quad 6 \quad 7 \quad 8 \quad j \quad 16 \quad k \quad k$$

where h, j and k are integers.

The median of the eight numbers is 10

The mode of the eight numbers is 18

The range of the eight numbers is 13

Work out the value of h, the value of j and the value of k

$h =$ ....

$j =$ ....

$k =$ ....

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The median is

The median is 10:

$$\frac{8+j}{2} = 10$$

$$j = 12$$

2. Calculate statistic

The mode is 18, so

$$k = 18$$

3. The range is

The range is 13:

$$18 - h = 13$$

$$h = 5$$

FINAL ANSWER

$h = 5,\ j = 12,\ k = 18.$

PAST PAPER QUESTION

Q#: 2 Marks: 4

TOPIC: **Graphing Inequalities**

Jun 2024 P2H - Jun 2024 | P2H

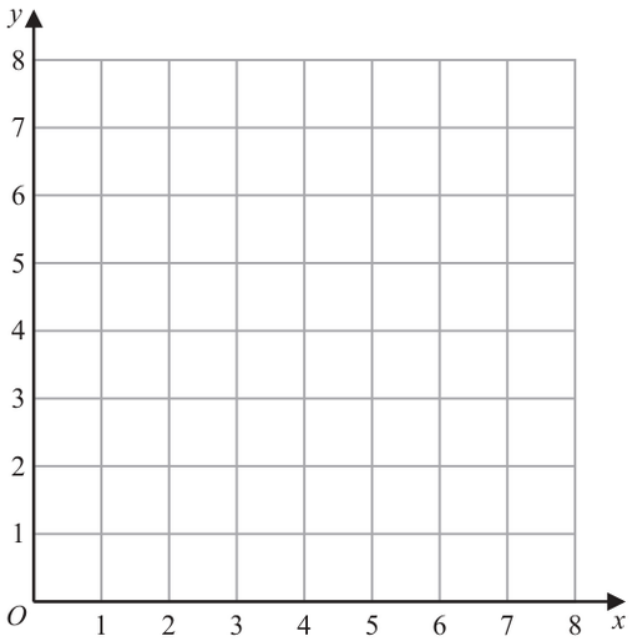
2 (a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$y \geqslant 2 \qquad x \leqslant 6 \qquad y \leqslant x + 1$

Label the region **R**

(1)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 4

Topic: **Graphing Inequalities**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Draw the boundary lines

Draw the boundary lines:

$$y = 2, \qquad x = 6, \qquad y = x + 1$$

2. The required region is

The required region is

$$y \geq 2$$

$$x \leq 6$$

$$y \leq x + 1$$

FINAL ANSWER

Shade $y \geq 2$, $x \leq 6$, $y \leq x + 1$.

PAST PAPER QUESTION

Q#: 3 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2024 P2H - Jun 2024 | P2H

3 A plane takes 9 hours 36 minutes to fly from New Delhi to Perth.

The plane flies at an average speed of 820 km/h.

Work out the total distance the plane flies.

.... km

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Evaluate fraction

$$9\text{ h }36\text{ min} = 9 + \frac{36}{60} = 9.6\text{ hours}$$

$$\text{distance} = 820 \times 9.6 = 7872$$

FINAL ANSWER
7872 km.

PAST PAPER QUESTION Q#: 4 Marks: 3

TOPIC: **Fractions**

Jun 2024 P2H - Jun 2024 | P2H

4 Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 3

TOPIC: **Fractions**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Simplify fraction

$$2\frac{4}{7} = \frac{18}{7}$$

$$3\frac{1}{9} = \frac{28}{9}$$

2. Simplify fraction

Now multiply:

$$\frac{18}{7} \times \frac{28}{9}$$

3. Cancel first

Cancel first:

$$\frac{18}{9} \times \frac{28}{7} = 2 \times 4 = 8$$

FINAL ANSWER

8

PAST PAPER QUESTION

Q#: 5 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2024 P2H - Jun 2024 | P2H

5 The diagram shows triangle *ABC*

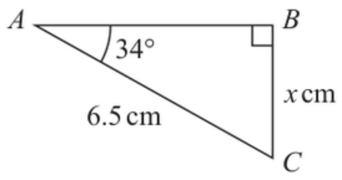


Diagram **NOT** accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$ ....

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

The hypotenuse is 6.5 cm, and x is opposite the $34°$ angle.

$$\sin 34° = \frac{x}{6.5}$$

$$x = 6.5 \sin 34° = 3.634\ldots$$

FINAL ANSWER

3.6 cm.

PAST PAPER QUESTION

Q#: 6 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2024 P2H - Jun 2024 | P2H

6 Change a speed of w metres per second to a speed in kilometres per hour.
Give your answer in terms of w in its simplest form.

.... kilometres per hour

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Convert m/s to km/h multiply

To convert m/s to km/h, multiply by 3.6.

$$w \text{ m/s} = 3.6w \text{ km/h}$$

$$3.6w = \frac{18w}{5}$$

FINAL ANSWER

$\frac{18w}{5}$ km/h.

PAST PAPER QUESTION

Q#: 7 Marks: 4

TOPIC: **Area & Perimeter**

Jun 2024 P2H - Jun 2024 | P2H

7 The diagram shows a 6-sided shape *ABCDEF*

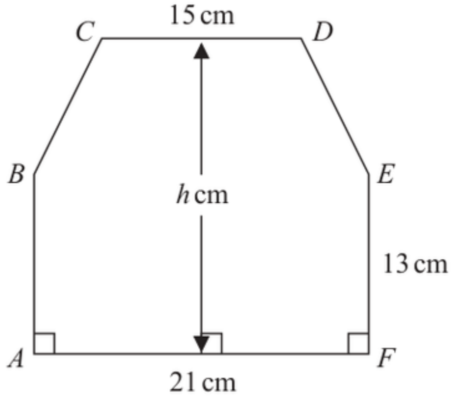


Diagram **NOT** accurately drawn

$AF = 21$ cm $\quad CD = 15$ cm $\quad AB = FE = 13$ cm

The perpendicular height of the shape is h cm
CD is parallel to AF

The area of the shape is 390 cm^2

Work out the value of h

$h =$ ....

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 4

TOPIC: **Area & Perimeter**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

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1. The bottom rectangle has area

The bottom rectangle has area

$$21 \times 13 = 273$$

2. The top trapezium has height

The top trapezium has height $h - 13$, with parallel sides 21 and 15.

$$273 + \frac{1}{2}(21 + 15)(h - 13) = 390$$

$$273 + 18(h - 13) = 390$$

$$18h + 39 = 390$$

$$h = 19.5$$

FINAL ANSWER

19.5 cm.

PAST PAPER QUESTION

TOPIC: **Percentages**

Jun 2024 P2H - Jun 2024 | P2H

Q#: 8 Marks: 5

8 Ishir plants 600 bulbs in a garden.
He plants tulip bulbs, crocus bulbs and daffodil bulbs so that

number of tulip bulbs : number of crocus bulbs : number of daffodil bulbs = 9 : 4 : 2

45% of the tulip bulbs are for yellow flowers.

$\frac{5}{8}$ of the crocus bulbs are for yellow flowers.

All of the daffodil bulbs are for yellow flowers.

Work out the number of bulbs that are for yellow flowers.

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 5

TOPIC: **Percentages**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The ratio parts are

The ratio parts are

$$9 + 4 + 2 = 15$$

$$\text{tulips} = 600 \times \frac{9}{15} = 360, \quad \text{crocuses} = 600 \times \frac{4}{15} = 160, \quad \text{daffodils} = 600 \times \frac{2}{15} = 80$$

2. Yellow tulips

Yellow tulips:

$$45\% \times 360 = 0.45 \times 360 = 162$$

3. Yellow crocuses

Yellow crocuses:

$$\frac{5}{8} \times 160 = 100$$

4. All the daffodil bulbs are

All the daffodil bulbs are yellow, so

$$162 + 100 + 80 = 342$$

FINAL ANSWER

342 bulbs

PAST PAPER QUESTION

Q#: 9 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2024 P2H - Jun 2024 | P2H

9 Giovanni invests 4500 koruna in a savings account for 4 years.
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of 4 years.
Give your answer correct to the nearest koruna.

.... koruna

(Total for Question 9 is 3 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The multiplier for compound interest

The multiplier for 2.4% compound interest is

$$1.024$$

2. Use compound interest

After 4 years,

$$4500(1.024)^4 = 4947.8023\ldots$$

3. Use compound interest

Correct to the nearest koruna,

$$4947.8023\ldots \approx 4948$$

FINAL ANSWER

4948 koruna

PAST PAPER QUESTION

Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

Jun 2024 P2H - Jun 2024 | P2H

10 Solve the simultaneous equations

$$6x + 4y = 1$$
$$3x + 5y = 8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION Q#: 10 Marks: 3

TOPIC: **Simultaneous Equations**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. Solve simultaneous equations

$$6x + 4y = 1$$

$$3x + 5y = 8$$

2. Double the second equation

Double the second equation:

$$6x + 10y = 16$$

3. Subtract the first equation

Subtract the first equation:

$$6y = 15$$

$$y = \frac{5}{2}$$

4. Solve simultaneous equations

Substitute into $3x + 5y = 8$:

$$3x + 5\left(\frac{5}{2}\right) = 8$$

$$3x + \frac{25}{2} = 8$$

$$3x = -\frac{9}{2}$$

$$x = -\frac{3}{2}$$

FINAL ANSWER

$x = -\frac{3}{2}$, $y = \frac{5}{2}$.

PAST PAPER QUESTION

Q#: 11 Marks: 3

TOPIC: **Factorising**

Jun 2024 P2H - Jun 2024 | P2H

11 (i) Factorise $x^2 + 9x - 22$

....

(2)

(ii) Hence, solve $x^2 + 9x - 22 = 0$

....

(1)

(Total for Question 11 is 3 marks)

PAST PAPER SOLUTION Q#: 11 Marks: 3

TOPIC: **Factorising**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. Factorise expression

$$x^2 + 9x - 22 = (x + 11)(x - 2)$$

2. Factorise expression

So

$$(x + 11)(x - 2) = 0$$

FINAL ANSWER

$x = -11$ or $x = 2$

PAST PAPER QUESTION

Q#: 12 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2024 P2H - Jun 2024 | P2H

12 Ali uses a fitness tracker to count the number of steps he walks each day for 7 days.

For the first 4 days, his mean number of steps is 11 800
For the next 3 days, his mean number of steps is 13 207

Work out his mean number of steps for the 7 days.

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

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1. Total for the first 4

Total for the first 4 days:

$$4 \times 11800 = 47200$$

2. Total for the next 3

Total for the next 3 days:

$$3 \times 13207 = 39621$$

3. Total for all 7 days

Total for all 7 days:

$$47200 + 39621 = 86821$$

4. Mean for the 7 days

Mean for the 7 days:

$$\frac{86821}{7} = 12403$$

FINAL ANSWER

12403.

PAST PAPER QUESTION

Q#: 13 Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

Jun 2024 P2H - Jun 2024 | P2H

13 The table gives information about the distances, in km, that 70 teachers travel to school.

Distance (d km)	Frequency
$0 < d \leqslant 10$	7
$10 < d \leqslant 20$	17
$20 < d \leqslant 30$	18
$30 < d \leqslant 40$	14
$40 < d \leqslant 50$	10
$50 < d \leqslant 60$	4

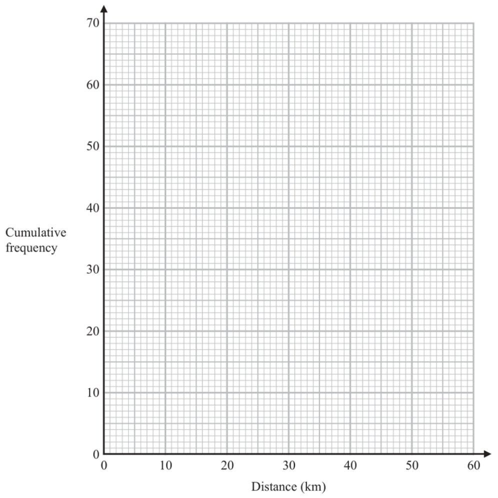
(a) Complete the cumulative frequency table.

Distance (d km)	Cumulative frequency
$0 < d \leqslant 10$	
$0 < d \leqslant 20$	
$0 < d \leqslant 30$	
$0 < d \leqslant 40$	
$0 < d \leqslant 50$	
$0 < d \leqslant 60$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the interquartile range of the distances.

.... km

(2)

(d) Use your graph to find an estimate for the number of teachers who travel more than 46 km.

....

(2)

(Total for Question 13 is 7 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 7

Topic: **Cumulative Frequency Diagrams**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The cumulative frequencies are

The cumulative frequencies are

$$7,\ 24,\ 42,\ 56,\ 66,\ 70$$

2. Plot these points and join

Plot these points and join them with a smooth curve:

$$(0,0),(10,7),(20,24),(30,42),(40,56),(50,66),(60,70)$$

3. Use cumulative frequency

For 70 teachers,

$$Q_1 = 17.5, \qquad Q_3 = 52.5$$

4. Use cumulative frequency

From the graph,

$$Q_1 \approx 16.2, \qquad Q_3 \approx 37.5$$

$$\text{IQR} \approx 37.5 - 16.2 = 21.3$$

5. Use cumulative frequency

At 46 km, the cumulative frequency is about 62.

$$70 - 62 = 8$$

FINAL ANSWER

IQR about 21 km, and about 8 teachers.

PAST PAPER QUESTION

Q#: 14 Marks: 7

TOPIC: **Algebraic Proof**

Jun 2024 P2H - Jun 2024 | P2H

14 (a) Show that $3y(2y + 5)(y + 7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers.

(3)

(b) Solve $\dfrac{2x+3}{5} + \dfrac{6x-5}{4} = \dfrac{163}{100}$

Show clear algebraic working.

$x =$ ....

(4)

(Total for Question 14 is 7 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 7

TOPIC: **Algebraic Proof**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. (a)

(a)

$$3y(2y+5)(y+7)$$

2. Expand brackets

First expand the two brackets:

$$(2y+5)(y+7) = 2y^2 + 14y + 5y + 35 = 2y^2 + 19y + 35$$

3. Multiply by

Now multiply by $3y$:

$$3y(2y^2 + 19y + 35) = 6y^3 + 57y^2 + 105y$$

4. Calculate value

So $a = 6$, $b = 57$, and $c = 105$.

5. (b)

(b)

$$\frac{2x+3}{5} + \frac{6x-5}{4} = \frac{163}{100}$$

6. Multiply by 100

Multiply by 100:

$$20(2x+3) + 25(6x-5) = 163$$

$$40x + 60 + 150x - 125 = 163$$

$$190x - 65 = 163$$

$$190x = 228$$

$$x = \frac{228}{190} = \frac{6}{5}$$

FINAL ANSWER

$6y^3 + 57y^2 + 105y$, and $x = \frac{6}{5}$.

PAST PAPER QUESTION

Q#: 15 Marks: 7

TOPIC: **Rearranging Formulas**

Jun 2024 P2H - Jun 2024 | P2H

15 (a) Make g the subject of $e = \sqrt{\dfrac{7g+5}{11+2g}}$

....

(4)

(b) Solve the inequality $3y^2 + 4y - 32 > 0$

Show your working clearly.

....

(3)

(Total for Question 15 is 7 marks)

PAST PAPER SOLUTION Q#: 15 Marks: 7

TOPIC: **Rearranging Formulas**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION *Dr. Eslam Ahmed*

1. (a)

(a)

$$e = \sqrt{\frac{7g+5}{11+2g}}$$

$$e^2 = \frac{7g+5}{11+2g}$$

$$e^2(11+2g) = 7g+5$$

$$g(2e^2-7) = 5-11e^2$$

$$g = \frac{5-11e^2}{2e^2-7}$$

2. (b)

(b)

$$3y^2+4y-32 = (3y-8)(y+4)$$

3. Solve quadratic equation

The critical values are $y=-4$ and $y=\frac{8}{3}$. Since the quadratic is positive outside the roots,

$$y < -4 \quad \text{or} \quad y > \frac{8}{3}$$

FINAL ANSWER

(a) $g = \frac{5-11e^2}{2e^2-7}$, (b) $y < -4$ or $y > \frac{8}{3}$

PAST PAPER QUESTION

Q#: 16 Marks: 7

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

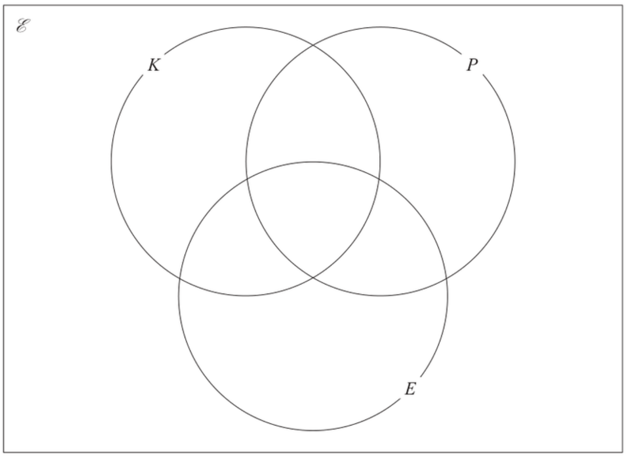
Jun 2024 P2H - Jun 2024 | P2H

16 60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

- 9 chose knitting, photography and embroidery
- 17 chose knitting and photography
- 16 chose photography and embroidery
- 20 chose knitting and embroidery
- 28 chose photography
- 39 chose embroidery
- 2 chose none of the workshops

(a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



(3)

One of the students is chosen at random.

Given that this student chose photography,

(b) find the probability that this student also chose knitting.

....

(2)

(c) Find $n(P \cap K')$

....

(1)

(d) Find $n([P \cup E] \cap K)$

....

(1)

(Total for Question 16 is 7 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 7

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The triple intersection is

The triple intersection is 9.

$$K \cap P \text{ only} = 17 - 9 = 8$$

$$P \cap E \text{ only} = 16 - 9 = 7$$

$$K \cap E \text{ only} = 20 - 9 = 11$$

$$P \text{ only} = 28 - 8 - 7 - 9 = 4$$

$$E \text{ only} = 39 - 11 - 7 - 9 = 12$$

2. There are students inside the

There are $60 - 2 = 58$ students inside the Venn diagram.

$$K \text{ only} = 58 - (8 + 7 + 11 + 9 + 4 + 12) = 7$$

$$P(K \mid P) = \frac{17}{28}$$

$$n(P \cap K') = 4 + 7 = 11$$

$$n((P \cup E) \cap K) = 8 + 11 + 9 = 28$$

FINAL ANSWER

$\frac{17}{28}$, 11, 28.

PAST PAPER QUESTION

TOPIC: **Direct & Inverse Proportion**

Jun 2024 P2H - Jun 2024 | P2H

Q#: 17 Marks: 3

17 Q is directly proportional to the square root of d

$Q = 4.5$ when $d = 324$

Find a formula for Q in terms of d

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Direct & Inverse Proportion**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Find inverse function

Q is directly proportional to $\sqrt{d}$, so

$$Q = k\sqrt{d}$$

2. Use when

Use $Q = 4.5$ when $d = 324$:

$$4.5 = k\sqrt{324}$$

$$4.5 = 18k$$

$$k = 0.25$$

3. Find inverse function

Therefore

$$Q = \frac{1}{4}\sqrt{d}$$

FINAL ANSWER

$Q = \frac{1}{4}\sqrt{d}$.

PAST PAPER QUESTION

Q#: 18 Marks: 2

TOPIC: **Linear Graphs ($y = mx + c$)**

Jun 2024 P2H - Jun 2024 | P2H

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

(Total for Question 18 is 2 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 2

TOPIC: **Linear Graphs (y = mx + c)**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The line is

The line is

$$5y + 2x = 7$$

$$5y = -2x + 7$$

$$y = -\frac{2}{5}x + \frac{7}{5}$$

2. Find the gradient

So the gradient of line P is $-\frac{2}{5}$.

3. A line perpendicular to P

A line perpendicular to P has gradient

$$\frac{5}{2}$$

FINAL ANSWER

$\frac{5}{2}$.

PAST PAPER QUESTION

Q#: 19 Marks: 3

Topic: **Rounding, Estimation & Bounds**

Jun 2024 P2H - Jun 2024 | P2H

19 $G = \dfrac{c}{2f - 3h}$

$c = 8$ correct to the nearest whole number
$f = 6.62$ correct to 2 decimal places
$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of G
Give your answer correct to 3 decimal places.
Show your working clearly.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

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1. Estimate the value

$$G = \frac{c}{2f - 3h}$$

2. Find lower bound

For the lower bound, use the lower bound of c and make the denominator as large as possible.

$$c \geq 7.5$$

$$f < 6.625, \quad h \geq 1.15$$

$$(2f - 3h)_{\text{upper}} = 2(6.625) - 3(1.15) = 9.8$$

$$G_{\text{lower}} = \frac{7.5}{9.8} = 0.765306\ldots$$

FINAL ANSWER

0.765

PAST PAPER QUESTION Q#: 20 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2024 P2H - Jun 2024 | P2H

20 Given that $k = x - y$ and $x = \dfrac{1}{4y}$

express $\dfrac{5k}{x+2}$ in the form $\dfrac{a - by^2}{c + dy}$ where a, b, c and d are integers.

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Simplify fraction

$$k = x - y, \quad x = \frac{1}{4y}$$

2. Simplify fraction

So

$$k = \frac{1}{4y} - y = \frac{1 - 4y^2}{4y}$$

3. Simplify fraction

Also

$$x + 2 = \frac{1}{4y} + 2 = \frac{1 + 8y}{4y}$$

4. Simplify fraction

Therefore

$$\frac{5k}{x+2} = \frac{5\left(\frac{1-4y^2}{4y}\right)}{\frac{1+8y}{4y}}$$

5. Cancel

Cancel $4y$:

$$\frac{5(1-4y^2)}{1+8y}$$

$$= \frac{5 - 20y^2}{1 + 8y}$$

FINAL ANSWER

$\dfrac{5 - 20y^2}{1 + 8y}$

PAST PAPER QUESTION Q#: 20 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2024 P2H - Jun 2024 | P2H

20 Given that $k = x - y$ and $x = \dfrac{1}{4y}$

express $\dfrac{5k}{x+2}$ in the form $\dfrac{a - by^2}{c + dy}$ where a, b, c and d are integers.

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Simplify fraction

$$k = x - y, \quad x = \frac{1}{4y}$$

2. Simplify fraction

So

$$k = \frac{1}{4y} - y = \frac{1 - 4y^2}{4y}$$

3. Simplify fraction

Also

$$x + 2 = \frac{1}{4y} + 2 = \frac{1 + 8y}{4y}$$

4. Simplify fraction

Therefore

$$\frac{5k}{x + 2} = \frac{5\left(\frac{1-4y^2}{4y}\right)}{\frac{1+8y}{4y}}$$

5. Cancel

Cancel $4y$:

$$\frac{5(1 - 4y^2)}{1 + 8y}$$

$$= \frac{5 - 20y^2}{1 + 8y}$$

FINAL ANSWER

$$\frac{5 - 20y^2}{1 + 8y}$$

PAST PAPER QUESTION

Q#: 21 Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Jun 2024 P2H - Jun 2024 | P2H

21 The diagram shows a square *ABCD* and a circle.

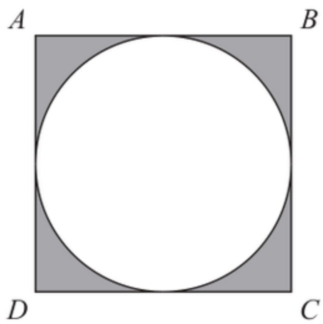

Diagram **NOT** accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is $80\,\text{cm}^2$

Work out the length of *AC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let the side length

Let the side length of the square be s.

2. Square

The circle is inscribed in the square, so its radius is

$$\frac{s}{2}$$

3. The shaded area is the

The shaded area is the area of the square minus the area of the circle:

$$s^2 - \pi\left(\frac{s}{2}\right)^2 = 80$$

$$s^2 - \frac{\pi s^2}{4} = 80$$

$$s^2\left(1 - \frac{\pi}{4}\right) = 80$$

$$s = \sqrt{\frac{80}{1 - \frac{\pi}{4}}} = 19.3076\ldots$$

4. The diagonal of the square

The diagonal of the square is

$$AC = s\sqrt{2}$$

$$AC = 19.3076\ldots\sqrt{2} = 27.3051\ldots$$

FINAL ANSWER

$\boxed{27.3 \text{ cm}}$

PAST PAPER QUESTION Q#: 21 Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Jun 2024 P2H - Jun 2024 | P2H

21 The diagram shows a square *ABCD* and a circle.

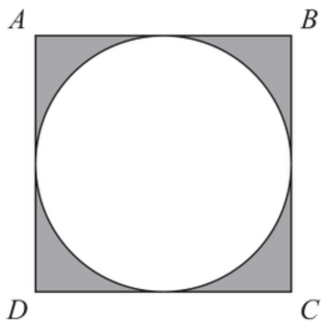

Diagram **NOT** accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is 80 cm^2

Work out the length of *AC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Let the side length

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4. The diagonal of the square

The diagonal of the square is

$$AC = s\sqrt{2}$$

$$AC = 19.3076\ldots\sqrt{2} = 27.3051\ldots$$

FINAL ANSWER

$\boxed{27.3 \text{ cm}}$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2024 P2H - Jun 2024 | P2H

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.................... , ...) (.................... , ...)

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The line is

The line is

$$x + y = 5$$

2. Calculate value

So

$$y = 5 - x$$

3. Substitute into the curve

Substitute into the curve:

$$2x^2 + 3y^2 = 210$$

$$2x^2 + 3(5 - x)^2 = 210$$

$$2x^2 + 3(x^2 - 10x + 25) = 210$$

$$5x^2 - 30x + 75 = 210$$

$$5x^2 - 30x - 135 = 0$$

4. Divide by 5

Divide by 5:

$$x^2 - 6x - 27 = 0$$

5. Factorise

Factorise:

$$(x - 9)(x + 3) = 0$$

6. Calculate value

So

$$x = 9 \quad \text{or} \quad x = -3$$

7. Calculate value

Using $y = 5 - x$:

$$x = 9 \Rightarrow y = -4$$

$$x = -3 \Rightarrow y = 8$$

FINAL ANSWER
$(9, -4)$ and $(-3, 8)$

PAST PAPER QUESTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2024 P2H - Jun 2024 | P2H

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.......................... , ...) (.......................... , ...)

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The line is

The line is

$$x + y = 5$$

2. Calculate value

So

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Substitute into the curve:

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Divide by 5:

$$x^2 - 6x - 27 = 0$$

5. Factorise

Factorise:

$$(x - 9)(x + 3) = 0$$

6. Calculate value

So

$$x = 9 \quad \text{or} \quad x = -3$$

7. Calculate value

Using $y = 5 - x$:

$$x = 9 \Rightarrow y = -4$$

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FINAL ANSWER
$(9, -4)$ and $(-3, 8)$

PAST PAPER QUESTION

Q#: 23 Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jun 2024 P2H - Jun 2024 | P2H

23 Simplify $\dfrac{30 \times 25^{2x+7}}{\sqrt{180} \times \left(\sqrt{5}\right)^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Simplify surd

$$\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$$

2. Simplify the number part

Simplify the number part:

$$\sqrt{180} = 6\sqrt{5}$$

$$\frac{30}{6\sqrt{5}} = \frac{5}{\sqrt{5}} = \sqrt{5} = 5^{1/2}$$

3. Simplify surd

Now write the powers with base 5:

$$25^{2x+7} = (5^2)^{2x+7} = 5^{4x+14}$$

$$(\sqrt{5})^{4x+9} = 5^{(4x+9)/2} = 5^{2x+9/2}$$

4. Use index laws

So the full expression is

$$5^{1/2+4x+14-(2x+9/2)}$$

$$= 5^{2x+10}$$

FINAL ANSWER

5^{2x+10}.

PAST PAPER QUESTION Q#: 23 Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jun 2024 P2H - Jun 2024 | P2H

23 Simplify $\dfrac{30 \times 25^{2x+7}}{\sqrt{180} \times \left(\sqrt{5}\right)^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

(Total for Question 23 is 3 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Simplify surd

$$\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$$

$$\frac{30}{\sqrt{180}} = \frac{30}{6\sqrt{5}} = \sqrt{5} = 5^{1/2}$$

$$25^{2x+7} = (5^2)^{2x+7} = 5^{4x+14}$$

$$(\sqrt{5})^{4x+9} = 5^{2x+\frac{9}{2}}$$

2. Evaluate fraction

So

$$5^{1/2} \times 5^{4x+14} \div 5^{2x+\frac{9}{2}} = 5^{\frac{1}{2}+4x+14-2x-\frac{9}{2}}$$

$$= 5^{2x+10}$$

FINAL ANSWER

5^{2x+10}, so $w = 2x + 10$

PAST PAPER QUESTION

Q#: 24 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2H - Jun 2024 | P2H

24

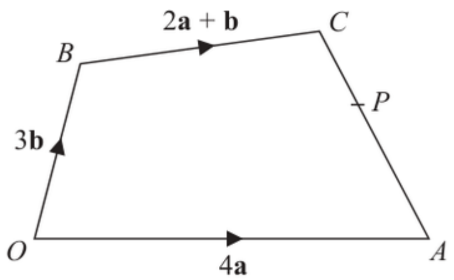


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$\overrightarrow{OA} = 4\mathbf{a}$ $\quad \overrightarrow{OB} = 3\mathbf{b}$ $\quad \overrightarrow{BC} = 2\mathbf{a} + \mathbf{b}$

(a) Find $\overrightarrow{AC}$ in terms of **a** and **b**
Give your answer in its simplest form.

$\overrightarrow{AC} =$ ....

(2)

The point P lies on AC such that $AP : PC = 3 : 2$

The point Q is such that OPQ and BCQ are straight lines.

(b) Using a vector method, find $\overrightarrow{OQ}$ in terms of **a** and **b**
Give your answer in its simplest form.
Show your working clearly.

$\overrightarrow{OQ} =$ ....

(4)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Read the graph

$$\vec{OA} = 4\mathbf{a}, \qquad \vec{OB} = 3\mathbf{b}, \qquad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

2. Read the graph

So

$$\vec{OC} = \vec{OB} + \vec{BC}$$

$$\vec{OC} = 2\mathbf{a} + 4\mathbf{b}$$

3. Read the graph

For part (a),

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$\vec{AC} = (2\mathbf{a} + 4\mathbf{b}) - 4\mathbf{a}$$

$$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}$$

4. Read the graph

For part (b), $AP : PC = 3 : 2$, so

$$\vec{OP} = \vec{OA} + \frac{3}{5}\vec{AC}$$

$$\vec{OP} = 4\mathbf{a} + \frac{3}{5}(4\mathbf{b} - 2\mathbf{a})$$

$$\vec{OP} = \frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

5. Read the graph

Since O, P, Q are collinear,

$$\vec{OQ} = t\vec{OP}$$

$$\vec{OQ} = t\left(\frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}\right)$$

6. Read the graph

Since B, C, Q are collinear,

$$\vec{OQ} = \vec{OB} + s\vec{BC}$$

$$\vec{OQ} = 3\mathbf{b} + s(2\mathbf{a} + \mathbf{b})$$

$$\vec{OQ} = 2s\mathbf{a} + (3 + s)\mathbf{b}$$

7. Equate coefficients

Equate coefficients:

$$\frac{14}{5}t = 2s$$

$$\frac{12}{5}t = 3 + s$$

8. Solve equation

From the first equation,

$$s = \frac{7}{5}t$$

9. Substitute into the second

Substitute into the second:

$$\frac{12}{5}t = 3 + \frac{7}{5}t$$

$$t = 3$$

10. Read the graph

Therefore

$$\vec{OQ} = 3\vec{OP}$$

$$\vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$$

FINAL ANSWER

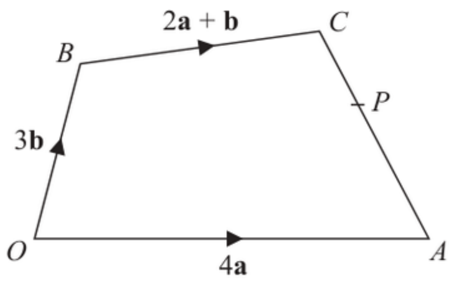
$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}$, $\vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$

PAST PAPER QUESTION Q#: 24 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2H - Jun 2024 | P2H

24



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Give your answer in its simplest form.

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Show your working clearly.

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(4)

(Total for Question 24 is 6 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Read the graph

$$\vec{OA} = 4\mathbf{a}, \qquad \vec{OB} = 3\mathbf{b}, \qquad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

2. Read the graph

So

$$\vec{OC} = \vec{OB} + \vec{BC}$$

$$\vec{OC} = 2\mathbf{a} + 4\mathbf{b}$$

3. Read the graph

For part (a),

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$\vec{AC} = (2\mathbf{a} + 4\mathbf{b}) - 4\mathbf{a}$$

$$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}$$

4. Read the graph

For part (b), $AP : PC = 3 : 2$, so

$$\vec{OP} = \vec{OA} + \frac{3}{5}\vec{AC}$$

$$\vec{OP} = 4\mathbf{a} + \frac{3}{5}(4\mathbf{b} - 2\mathbf{a})$$

$$\vec{OP} = \frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

5. Read the graph

Since O, P, Q are collinear,

$$\vec{OQ} = t\vec{OP}$$

$$\vec{OQ} = t\left(\frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}\right)$$

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Since B, C, Q are collinear,

$$\vec{OQ} = \vec{OB} + s\vec{BC}$$

$$\vec{OQ} = 3\mathbf{b} + s(2\mathbf{a} + \mathbf{b})$$

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Substitute into the second:

$$\frac{12}{5}t = 3 + \frac{7}{5}t$$

$$t = 3$$

10. Read the graph

Therefore

$$\vec{OQ} = 3\vec{OP}$$

$$\vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$$

FINAL ANSWER

$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}$, $\vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$

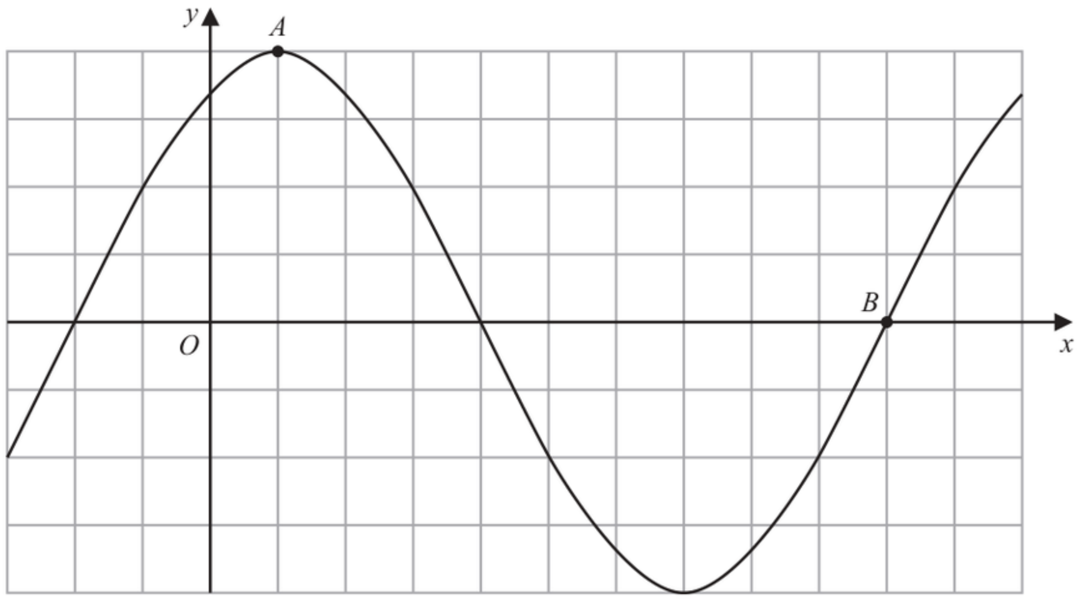
PAST PAPER QUESTION

Q#: 25 Marks: 2

TOPIC: **Graphs of Functions**

Jun 2024 P2H - Jun 2024 | P2H

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.................... , ...)

(1)

(ii) Find the coordinates of the point B

(.................... , ...)

(1)

(Total for Question 25 is 2 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 2

TOPIC: **Graphs of Functions**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometry

$$y = 2\sin(x+60)^\circ$$

2. Point is the maximum point

Point A is the maximum point.

3. The maximum value is

The maximum value is

$$y = 2$$

4. This happens when

This happens when

$$x + 60 = 90$$

$$x = 30$$

5. Read the graph

So

$$A = (30, 2)$$

6. Point is the positive intercept

Point B is the positive x-intercept after the minimum.

$$2\sin(x+60)^\circ = 0$$

$$\sin(x+60)^\circ = 0$$

7. Read the graph

For this point,

$$x + 60 = 360$$

$$x = 300$$

8. Read the graph

So

$$B = (300, 0)$$

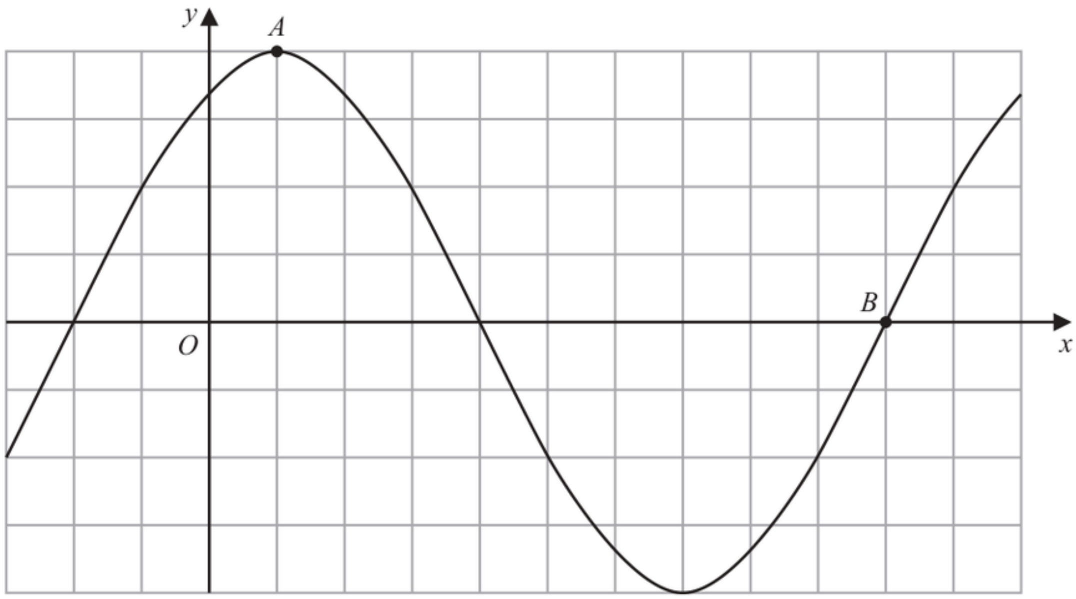
FINAL ANSWER
$A = (30, 2),\ B = (300, 0)$

PAST PAPER QUESTION Q#: 25 Marks: 2

TOPIC: **Graphs of Functions**

Jun 2024 P2H - Jun 2024 | P2H

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.................... , ...)

(1)

(ii) Find the coordinates of the point B

(.................... , ...)

(1)

(Total for Question 25 is 2 marks)

PAST PAPER SOLUTION

Q#: 25 Marks: 2

TOPIC: **Graphs of Functions**

Jun 2024 P2H - Jun 2024 | P2H

WORKED SOLUTION

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1. Use trigonometry

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Point A is the maximum point.

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Point B is the positive x-intercept after the minimum.

$$2\sin(x + 60)^\circ = 0$$

$$\sin(x + 60)^\circ = 0$$

7. Read the graph

For this point,

$$x + 60 = 360$$

$$x = 300$$

8. Read the graph

So

$$B = (300, 0)$$

FINAL ANSWER
$A = (30, 2),\ B = (300, 0)$